Task 3 - Four-Bar Linkage Motion Simulation

Introduction

The purpose of this task was to build a simple mechanism in FreeCAD and study its motion. A four-bar linkage was selected because it clearly shows how rotary motion from an input crank is transferred through a coupler to an output rocker. The model was then used to check the motion, velocity, acceleration and forces during one complete crank cycle.

Model Setup

The mechanism was created with four links connected by revolute joints. The fixed ground link is 120 mm long. The input crank is 40 mm, the coupler is 100 mm and the output rocker is 80 mm. The input crank was set to rotate at 60 rpm. A motion envelope with several crank positions was also included so the movement of the linkage can be seen clearly.

Motion Simulation

The crank was rotated through a full 360 degree cycle. As the crank turns, the coupler transfers the motion to the rocker. The rocker does not rotate continuously; it moves back and forth between two limiting positions. The joint paths and the different linkage positions were used to confirm that the mechanism moves smoothly without the links separating.

Results and Discussion

The simulation produced graphs for output rocker velocity, output rocker acceleration, joint force and required input torque. The largest absolute rocker velocity was about 3.83 rad/s and the largest absolute rocker acceleration was about 35.35 rad/s^2. The maximum dynamic force at joint C was about 0.756 N, while the peak input torque was about 0.0321 N m. The graphs show that these values change throughout the cycle because the linkage geometry is continuously changing. The highest acceleration and force occur when the direction or speed of the rocker changes more quickly.

Conclusion

The FreeCAD model successfully demonstrated the motion of a four-bar linkage. The simulation showed the movement of the links and provided the required velocity, acceleration and force results. The motion envelope and result graphs made it easier to understand how the mechanism behaves during a full crank rotation. Overall, the project met the requirements of Task 3 and showed how motion simulation can be used to study a mechanical linkage before it is built physically.